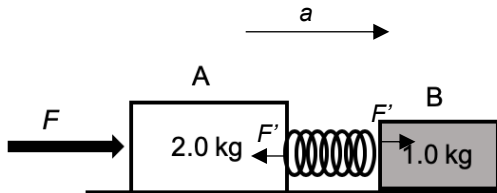


$$10 = u(0.875) + \frac{1}{2}(9.81)(0.875)^2 \Rightarrow u = 7.1 \text{ m s}^{-1}$$

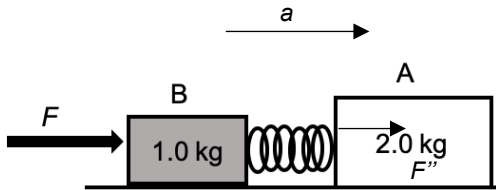
5 A



$$F = ma = (2.0 + 1.0) a$$

$$a = \frac{F}{3}$$

$$F' = kx_1 = \frac{F}{3} \Rightarrow x_1 = \frac{F}{3k}$$



$$F'' = \frac{F}{3} \times 2$$

$$kx_2 = \frac{2F}{3} \Rightarrow x_2 = \frac{2F}{3k} = 2x_1$$

$$\Rightarrow x_1 : x_2 = 1 : 2$$